

EMMANUEL N MAKOKHA

Software Engineer | AI/ ML
Engineer

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PROFILE SUMMARY

Results-driven Senior Software Engineer with 6+ years of experience specializing in Python and advanced Machine Learning. Expertise in architecting, training, fine-tuning, and deploying Large Language Models (LLMs) using Hugging Face Transformers and local AI tools like Ollama. Proven track record in Retrieval-Augmented Generation (RAG), Agentic RAG, Hybrid RAG, AI agents with protocols (MCPs & A2A), LLM chaining, and prompt engineering for production environments. Skilled in validating LLM outputs with real-world datasets, integrating AI with secure Python backends (Flask, Django), and deploying scalable solutions on AWS, Azure, GCP. Published articles on agentic AI on Medium and contributed to open-source frameworks on GitHub.

EDUCATION

Multimedia University of Kenya

BSc. in Software Engineering

- From 2015 September to 2019 October. -
Graduated with 2nd class honors.

St. Joseph's Boys School Kitale

Graduated with KCSE certificate 2014

- Excellence with A- mean grade.

CAREER BACKGROUND

Python, Machine Learning & AI Development (2024 July- 2025)

- Built and deployed LLM-powered applications focused on language understanding, RAG, Agentic workflows, and prompt engineering, enhancing response quality and automation.
- Trained and fine-tuned LLMs using Hugging Face Transformers and Ollama for local AI solutions; implemented validation pipelines with open-source datasets.
- Developed AI agents using Python, Pydantic, and LangGraph; published articles on Medium (e.g., "The AI Protocol Wars: MCP vs. A2A" and "Building Structured AI Agents") with practical implementations on GitHub.
- Integrated AI/ML with Python backends (Flask/Django) and cloud infrastructures (AWS/Azure/GCP) for production-grade features.
- Automated data flows and QA using agentic AI, Python scripting, and cloud functions in Agile teams.
- Integrated AI/ML services (LLMs, RAG systems, AI agents) with cloud-native backends, delivering intelligent automation solutions that reduced manual processing time by 40%.
- Built Flutter and web frontends consuming AI-powered APIs for automated, intelligent user experiences.
- Automated data flows, QA, and business processes using agentic AI, Python scripting, and cloud functions.

Python Backend Engineer & Data Science Path

From July 2023 to December 2024.

- Built and deployed scalable cloud infrastructures on AWS, Azure, and GCP, supporting 70% of production workloads with 99.9% uptime SLA achievement
- Architected microservices-based applications using Flask and Django, deployed on containerized environments with CI/CD pipelines for automated testing and deployment

- Developed validation frameworks to evaluate and optimize AI model outputs, ensuring production-grade quality and performance standards
- Automated infrastructure provisioning and configuration management using Infrastructure as Code (IaC) principles and DevOps best practices
- Implemented comprehensive monitoring and logging solutions for cloud applications, enabling proactive issue detection and resolution
- Built Flutter-based frontends consuming cloud-hosted APIs, delivering intelligent user experiences across cross-platform mobile applications.

TECHNICAL SKILLS

- AI/ML Development: LLM training/fine-tuning/inference (Hugging Face, PyTorch, TensorFlow); RAG/Agentic RAG/Hybrid RAG; AI agents & MCPs/A2A protocols; LLM chaining/workflows; Prompt engineering; LangChain/LangGraph/LlamaIndex.
- Local AI: Ollama, on-premise model deployment & evaluation; AI/LLM guardrails.
- Published technical articles on Medium covering agentic AI, Model Context Protocol (MCP), and A2A agent interoperability, such as "The AI Protocol Wars: A Comprehensive Analysis of MCP vs. A2A with Practical Implementations."
- Python Ecosystem: Flask, Django, Pandas, NumPy, Scikit-learn.
- Cloud & APIs: AWS, Azure, GCP; RESTful integration; CI/CD pipelines.
- Mobile/Web: Flutter (Dart), Firebase, WebSockets.
- Automation: Selenium, Playwright, UIPath; Web scraping with AI agents.
- Other: Git, Bitbucket, Docker.

Python, Cloud, Machine Learning (2023)

- **Advanced Automation and Flutter Integration**
 - Increased focus on automation systems and integration with production-level Flutter applications, achieving streamlined cross-platform solutions.
 - Deepened Python expertise, leveraging new Google certification in developer training to enhance and professionalize automation workflows.
- **Machine Learning and Cloud Solutions**
 - Intelligent autoscaling based on traffic spikes and historical patterns.
 - Self-healing Infrastructure by detecting anomalies in logs.
 - Serverless AI - Inference using Lambda and cloud functions
- **Multi-agent Systems (Emerging)**
 - Early exploration of agentic concepts and automation protocols in AI.

Python Backend, Web automation & Mobile Development

From July 2021 to December 2022.

- Developed Python-based automation scripts and ML pipelines, including early LLM explorations and agentic concepts.
- Built cross-platform Flutter applications consuming AI APIs; automated processes with UIPath, Selenium, and Playwright.
- Contributed to GitHub repositories for automation and data workflows.
- Managed Python backend development with Flask/Django; integrated Firebase/SQLite databases and RESTful APIs.
- Collaborated on UX designs and location-based features using Google Maps APIs.
- Orchestrated Flutter and Dart solutions for diverse projects for cross-platform applications.
- Automated processes utilizing Python in conjunction with UIPath, Selenium and the PlayWright Framework

TECHNICAL SKILLS CONT

- AI/ML & Data Engineering: LLM Training & Fine-tuning, RAG Systems, AI Agents, Huggingface Transformers, TensorFlow, PyTorch, Pandas, NumPy, Scikit-learn, Model Deployment & Optimization
- Cloud Security & Compliance: Cloud Security Best Practices, Identity & Access Management (IAM), Network Security, Data Protection, Compliance Standards
- Additional Technologies: Flutter/Dart, Firebase, Git, Bitbucket, Selenium, Playwright, Test Automation

CERTIFICATIONS

- **BSc.** Software Engineering from the Multimedia University of Kenya.
- Google Africa Developer Training Program - Advanced Cloud & Development Track
- Foundation for Test Automation (Test Automation University)
- Python Programming (Test Automation University)
- Source Control for Test Automation (Test Automation University)
- Data Science
- Automated testing by Test Automation University

Mobile Developer & Backend Systems Engineer -2021

- Developed Python backends for mobile applications, integrating cloud services and APIs for enhanced functionality
- Implemented Firebase Database solutions for real-time data synchronization and cloud-based storage
- Built cross-platform mobile applications using Flutter and Dart, deployed to cloud-hosted backend services
- Integrated Google Maps APIs and other cloud-based services for location-based features
- Collaborated with UX designers to implement cloud-connected features with optimal performance.

Mobile Developer

From May 2020 to March 2021

- Utilized Python for backend mobile application development, contributing to the creation of various high-functioning apps.
- Flutter and Dart for cross-platform and Firebase Database.
- Leveraged Google Maps APIs for location-based features in mobile apps.
- Collaborated with UX designers to convert wireframes and prototypes into functional and aesthetically pleasing designs.

LANGUAGES

- English
- Kiswahili

REFERENCES

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